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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular/Supplementary Examinations February/March-2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Artificial Intelligence Integrated Software Engineering

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

14/20

Question No	Question	M	CO	BT
1.1	What is incremental development? -4	01	1	1
1.2	Define software evolution.	01	1	1
1.3	What is requirements validation?	02	1	2
1.4	Differentiate between closed and open interviews.	02	1	2
1.5	Name the four essential elements of a design pattern.	02	1	2
1.6	What do you mean by legacy Systems?	01	2	1
1.7	What is Model-Driven Architecture (MDA)?	01	1	2
1.8	What is software design and implementation?	02	1	1
1.9	What is the purpose of feature extraction in ML-based code review?	02	4	2
1.10	What is the main drawback of static code analysis in CI?	02	4	2
1.11	What is Test Suite Reduction?	02	4	2
1.12	What is the purpose of mutation testing?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Illustrate the Organization of the Mentcare system. -4	06	1	2
2b	Illustrate the following i) Insulin Pump Hardware Architecture -3 -6 ii) Activity Model of the Insulin Pump -3	10	1	3
3a	Discuss the reason behind Requirements Changes in Software Engineering.	08	2	2

-10 6/16

3b	A user switches on the microwave oven and observes that the display initially shows the current time. The user then selects a power level, either Full Power (600 W) or Half Power (300 W), and sets the required cooking time using the timer. After setting the time, the user closes the microwave door and presses the Start button to begin cooking. During the operation, the user may open the door or press the Cancel button. Based on the given state diagram of the microwave oven, explain the sequence of states the system transitions through from the initial waiting state to the operation state and back to the waiting state. Describe the actions performed in each state, the events that trigger state transitions, and how safety conditions such as door opening are handled by the system.	08	2	4
	OR			
4a	Explain the Requirements Engineering Process with neat diagram. Discuss each activity in detail.	10	2	2
4b	Summarize the Client-Server architectural pattern by consideing an example.	06	2	4
5a	A Library Management System has a module called Book_Loan that allows users to borrow books. The system enforces the following rules: i) A user can borrow a book only if the book is available. ii)A user cannot borrow more than 3 books at a time. iii)The system should update the number of available copies after a successful loan. iv)If any rule is violated, the system should display an appropriate error message.	08	2	3
5b	Discuss the components of a design pattern with an example.	08	1	2
	OR			
6a	A company is developing an Online Examination System for a university. The system includes modules for student registration, exam scheduling, question paper delivery, online answer submission, result processing, and report generation. The software must be reliable, secure, and able to handle thousands of students simultaneously. Before deploying the system, the development team plans to follow a systematic software testing process to ensure quality and correctness. Discuss how the Software Testing Process should be applied to the Online Examination System described above.	10	2	2
6b	Explain how Configuration Management helps in managing multiple versions of software components and supports team collaboration.	06	2	2
7a	Explain the role of feature extraction (bag-of-words and sentiment analysis) in the ML system and justify why these techniques are effective for code review automation.	10	3	3
7b	Summarize the processes of Model Development and Model Validation in the AI-based MBSE framework.	06	3	2
	OR			
8a	Use the three-tank system as an example, to describe how process control is applied to maintain the tank levels in a Cyber-Physical System (CPS).	10	3	3
8b	A software development team initially performs manual code reviews, including review speed measurement, rule definition, code analysis, and feedback. To improve efficiency, the team gradually introduces	06	3	3

	automation, progressing from automated review speed measurement to automated violation analysis, recommendations, and finally fully automated code reviews. explain how stairway model improves code review efficiency and quality.			
9a	Discuss the role of Artificial Neural Networks and Genetic Algorithms in Test Case Generation.	10	4	3
9b	Describe the role of Artificial Neural Networks (ANN) in test-oracle automation.	06	4	3
	OR			
10a	Use of Hybrid Machine Learning models in software testing improve prediction and testing outcomes. Justify your answer.	08	4	4
10b	How clustering techniques in used in fault prediction and test suite optimization.	08	4	3